Alfredo Navarrete Montes

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EDUCATION

Benemerita Universidad Autonoma de Puebla (BUAP) Puebla, Mexico
Bachelor of Engineering in Information Technologies. *Expected Graduation: December 2026*
*Relevant Coursework (completed at **University of New Mexico, USA** - revalidated):* High Performance Computing, Information Systems Development, Computer-based Information Systems, Business Programming Fundamentals, Design of Large Programs.

EXPERIENCE

Volkswagen de Mexico (VWM) - IT Solutions Puebla, Mexico
Software Engineering Intern *August 2026 - Present*

- Developed in PowerShell and VBScript a scanner that identifies CBD files by content, recognizing multiple template versions in English, Spanish and German, converts legacy .xls files to .xlsx and uploads them to AWS S3.
- Processed .xlsx files by reading the underlying XML directly, without opening Excel, and .xls files through Excel COM, so scans never take over the user's Excel session.
- Designed a two-flow architecture: one scans the user profile without touching the originals, the other reviews files arriving through Power Automate (OneDrive for Business to SharePoint). Implemented incremental state control with file fingerprints (modified date and size) to skip processed files and version those that change.
- Built a single-file VBS installer that detects OneDrive, creates the folder structure and registers a silent daily task, working around GPO restrictions with PowerShell execution via stdin and schtasks XML.
- Built the SharePoint-to-AWS S3 upload stage using AWS CLI, working through the Zscaler corporate proxy, with rotating logs and failure-only alerts.
- Migrated authentication from a provisional technical user to IAM Roles Anywhere with a VW CA certificate, coordinating issuance with the PKI team, and validated the upload in QAS and production. Coordinated with the NTT DATA team (S3 consumers) and the AWS account administrator on naming, prefixes and permissions.

Health Technologies Laboratory - BUAP Puebla, Mexico
Research Assistant *January 2026 - August 2026*

- Participated in MentalRiskES 2026 (IberLEF 2026 shared task) on early detection of mental health risk from Spanish social media text, as part of the BUAP research team.
- Achieved 6th place overall in Task 1 and the best early-detection score (R10, MAE 0.889) with the lowest energy consumption among all participating teams.
- Built and ran the competition pipeline in Python, orchestrating multiple LLMs (GPT-4o-mini, GPT-4.1-nano, GPT-5-nano) through the OpenAI API with mock and dry-run workflows.
- Co-authored the resulting paper, accepted for publication in CEUR-WS (indexed in Scopus and DBLP).

University of New Mexico (UNM) - USA Albuquerque, NM
Research Assistant *August 2025 - December 2025*

- Designed a distributed Python scraping pipeline that collected 150k+ Kickstarter campaign records, enabling large-scale analysis of customer behavior.
- Developed automated scrapers in Python to extract structured and unstructured data from multiple online sources.
- Built data preprocessing workflows to clean and merge datasets for analysis and machine learning experiments.
- Collaborated with graduate researchers at the Anderson School of Management to analyze customer behavior and market trends.

HIGH PERFORMANCE COMPUTING PROJECTS

HPC Benchmarking - Center for Advanced Research Computing (UNM) *August 2025 - December 2025*

- Benchmarked HPL on the Hopper cluster (32 cores per node, InfiniBand, GCC), tuning matrix size N, block size NB, and process grid P x Q through automated Slurm parameter sweeps.
- Raised throughput from 209 to 1,106 GFLOP/s at a fixed rank count by optimizing process grid shape, and reached 2.95 TFLOP/s on 128 MPI ranks.
- Ran CPU-only HPCG experiments on the Easley cluster, scaling from 1 to 128 cores across 4 nodes and analyzing serial fraction and parallel efficiency for a GPU vs CPU study.
- Conducted strong and weak scaling analysis (Amdahl and Gustafson) and co-authored two technical reports in ACM format.

TECHNICAL SKILLS

Programming: Python, C++, TypeScript, PowerShell, VBScript
Distributed Systems & HPC: Slurm, MPI, HPL, HPCG, Linux
Cloud & Automation: AWS S3, AWS CLI, IAM Roles Anywhere, Power Automate, SharePoint, Task Scheduler
Web Development: Angular, Django, REST APIs, MySQL
Data & Machine Learning: NLP, Transformers, LLM APIs, Data Preprocessing, Embeddings
Tools: Git, GitHub

LANGUAGES

Spanish: Native | **English:** Advanced (TOEFL ITP: 560, Duolingo English Test: 120)